

Class Test 1: CS366	Duration: 30m Max-Marks: 25 Date: 28/01/2025	Roll: Name:
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[Q1] (2 marks - MSQ) Select the evaluation function(s) that will result in identical behaviour to greedy best-first search:

- ☐ [A] $f(n) = h(n)/2$
☐ [B] $f(n) = h(n)$
☐ [C] $f(n) = \sqrt{h(n)}$
☐ [D] $f(n) = g(n) \times h(n)$

[Q2] (2 marks - MCQ) Assume a uniform-cost-search (UCS) on a tree with branching factor b , if the cost function $g(n)$ is linear, $g(n) = k \times d(n)$ where k is a constant and $d(n)$ is the depth of node n , how many nodes are expanded by UCS before it finds the optimal solution at depth δ starting from the root-node?

- ☐ [A] at least: $(k\delta - 1)^k / (b - 1)$
☐ [B] at least: $(b^\delta - 1) / (b - 1)$
☐ [C] at most: $b(\delta^b - 1) / (b - 1)$
☐ [D] at most: $(b\delta - 1)^b / (b - 1)$

[Q3] (2 marks - MSQ) Select the correct statement(s) with regards to the minimax search algorithm:

- ☐ [A] It is optimal for all kinds of adversaries
☐ [B] It is complete if the search tree is finite
☐ [C] with pruning, it can potentially lead to suboptimal solutions
☐ [D] the order in which nodes are visited affects the amount of pruning

[Q4] (2 marks - MSQ) In minimax search with pruning, consider a player who had a chance of moving to a node n somewhere in the search tree, the player will choose not to move to this node when

- ☐ [A] It has a better choice at the same depth
☐ [B] It had a better choice at some lower depth
☐ [C] It has a better choice at some higher depth
☐ [D] If the other player has a better choice at the same depth

[Q5] (2 marks - MSQ) Which search methods use problem specific knowledge beyond the definition of the problem?

- ☐ [A] Depth First Search
☐ [B] A* Search
☐ [C] Greedy Best-First Search
☐ [D] Uniform Cost Search

[Q6] (2 marks - MSQ) Select is the correct statements for Depth-First Search (DFS)?

- ☐ [A] It is not guaranteed to find a solution even if one exists
☐ [B] It has linear time complexity
☐ [C] It does not guarantee optimality
☐ [D] It has better space complexity than BFS

[Q7] (2 mark - MSQ) Breadth-First Search (BFS) is optimal if:

- ☐ [A] the path cost is a non-decreasing function of the depth of the node
☐ [B] the path cost is a linear function of depth of the node
☐ [C] all actions/edges have the same cost
☐ [D] the search tree is finite

[Q8] (1 marks - MCQ) In the context of game trees, the "evaluation function" is primarily used to:

- ☐ [A] Prune the tree branches.
☐ [B] Estimate the utility of a non-terminal state.
☐ [C] Optimize the search depth.
☐ [D] Generate random moves for the opponent.

[Q9] (2 marks - MSQ) Which of the following are true about Alpha-Beta Pruning?

- ☐ [A] It can reduce the number of nodes evaluated by Minimax.
☐ [B] It guarantees the same result as Minimax without pruning.
☐ [C] It works only when the game tree is fully explored.
☐ [D] It eliminates the need to evaluate certain branches of the tree.

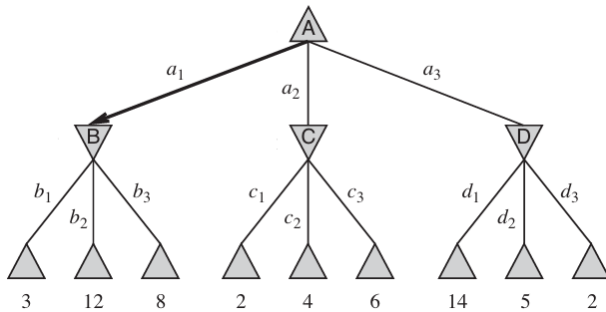
[Q10] (2 marks - MCQ) Consider an infinite grid of 2-D cells where each cell is represented by its integer coordinates. The agent starts at cell (0, 0) and can move in four orthogonal directions by just one step per move. The branching factor for such a search tree is ____ and the maximum depth of the search tree is ____.

- ☐ [A] 4, ∞
- ☐ [B] 3, ∞
- ☐ [C] ∞ , 4
- ☐ [D] ∞ , 3

[Q11] (3 marks - MCQ) Assume the same setup as in the previous question, the maximum number of nodes expanded by BFS given that the goal is located at a cell (100, -100) is:

- ☐ [A] 80400 ☐ [B] 20200
- ☐ [C] 90600 ☐ [D] 40200

For the following questions (12, 13 and 14), consider the 2-ply game as shown below:



[Q12] (1 marks - MCQ) The Minimax Values for nodes are:

- ☐ [A] A=14, B=12, C=6, D=14 ☐ [B] A=6, B=12, C=6, D=14
- ☐ [C] A=3, B=3, C=2, D=2 ☐ [D] A=2, B=3, C=2, D=2

[Q13] (1 marks - MCQ) With Alpha-Beta Pruning, what will be the final values of alpha and beta at node C?

- ☐ [A] $-\infty$, 6 ☐ [B] $-\infty$, 2 ☐ [C] ∞ , 4 ☐ [D] ∞ , 6

[Q14] (1 marks - MCQ) With Alpha-Beta Pruning, which of the following leaf nodes will be pruned?

- ☐ [A] c_2, c_3 ☐ [B] d_1, c_3
- ☐ [C] b_2, b_3 ☐ [D] c_2, b_2